

Social Engine

Exploratory Data Analysis — Analytical Commentary

Interpretation of the 20 analytical figures retained from the supplied EDA report. The commentary focuses on observed patterns, analytical significance, and limitations rather than describing the chart mechanics.

Dataset context: 12,000 cleaned posts from 1,500 users covering 01 May 2024 to 30 April 2025.

Interpretation note: Correlation values describe linear association only and should not be interpreted as evidence of causation.

1. Post Volume by Platform

The platform distribution is remarkably balanced among the five identified platforms. Facebook records the highest volume with 2,074 posts, followed closely by YouTube (2,073), Twitter (2,049), Reddit (2,031), and Instagram (1,989).

The difference between the most and least represented known platforms is only 85 posts, indicating that no single platform dominates the dataset. This provides a relatively balanced basis for cross-platform comparison.

However, 1,784 posts have no recorded platform. This is a substantial missing group and should remain explicitly unidentified rather than being assigned to a platform without evidence.

Platform-specific conclusions therefore apply only to posts with a known platform.

Finding: The identified platforms are highly balanced, but the unknown-platform group is a significant limitation for platform-level analysis.

2. Daily Posting Activity

Daily posting volume remains relatively stable throughout the one-year observation period. Most days fall within a fairly narrow range, with short-lived peaks and dips but no sustained upward or downward trend.

The isolated spikes are better interpreted as individual high-activity days rather than evidence of a broader temporal trend. The dataset therefore does not show a clear seasonal or long-term change in posting intensity.

Finding: Posting activity is comparatively stationary over the year, with variation driven mainly by short-term fluctuations.

3. Posting Activity by Day of Week

Posting activity varies only modestly across the seven days. Wednesday has the highest volume with 1,771 posts, while Saturday has the lowest with 1,675 posts.

The difference between these values is only 96 posts, so the weekly distribution is relatively even. There is no strong weekday-versus-weekend separation in the observed data.

Finding: Wednesday is the busiest day, but day-of-week has only a limited effect on total posting volume.

4. Posting Activity by Hour

Hourly activity is distributed across the full day rather than concentrated within a narrow period. The busiest known posting hour is 23:00 with 388 posts, while activity is somewhat lower around parts of the afternoon.

The differences between hourly counts are relatively modest. The late-evening maximum therefore represents a tendency rather than a dominant posting window.

Only posts with known posting times are included in this analysis. Date-only source timestamps do not provide enough information to infer an hour.

Finding: Posting occurs throughout the day, with a modest tendency toward higher activity during late evening hours.

5. Distribution of Likes

Likes are distributed broadly across approximately 0–5,000 rather than being concentrated near the lower end of the range. The mean is 2,491.86 and the median is 2,498.

The near-identical mean and median indicate limited aggregate skew. The distribution therefore does not appear to be dominated by a small number of extremely high-like posts.

Finding: Like counts are broadly distributed and comparatively balanced, with little evidence of strong aggregate skew.

6. Distribution of Shares

Shares are spread across approximately 0–2,500, with observations distributed throughout much of the range. The mean is 1,007.17 and the median is 1,018.

The close agreement between the mean and median indicates that extreme observations do not substantially distort the central tendency.

Finding: Sharing activity is widely distributed and does not exhibit strong aggregate skew.

7. Distribution of Comments

Comments range from near zero to approximately 1,000 per post. The mean is 504.35 and the median is 503.

The almost exact agreement between the two measures indicates that the overall distribution is comparatively balanced rather than being driven primarily by a small set of highly commented posts.

Finding: Comment activity is broadly distributed, with limited evidence of aggregate skew.

8. Likes by Platform

The platform-level boxplots show substantial overlap in the distributions of likes. Median values and interquartile ranges are broadly similar across Reddit, Facebook, Twitter, Instagram, and YouTube.

Although some platforms show slightly higher or lower central tendencies, those differences are small relative to the within-platform spread. Platform membership therefore does not appear to create a large separation in post-level likes.

Finding: Like distributions are broadly comparable across platforms, with within-platform variation larger than the visible differences between platforms.

9. Shares by Platform

Share distributions are highly similar across platforms. Instagram shows a somewhat higher median than several other platforms, while Facebook and Reddit are somewhat lower.

The substantial overlap between the distributions limits the strength of any platform-level distinction. The observed difference in Instagram's central tendency should therefore be treated as a modest pattern rather than a decisive platform effect.

Finding: Instagram has a somewhat higher typical share count, but platform-level distributions remain strongly overlapping.

10. Comments by Platform

The comment distributions are particularly similar across the five platforms. Median comment counts and overall spread remain close from one platform to another.

There is no visually dominant platform with a substantially different commenting profile. The data therefore provides little evidence that platform membership strongly affects comment volume.

Finding: Commenting behaviour is highly consistent across platforms.

11. Likes vs Shares

The scatter plot forms a dense cloud without a visible upward or downward pattern. Posts with many likes are not systematically associated with many shares.

The Pearson correlation is approximately -0.00, confirming that the two measures have essentially no linear association in this dataset.

Finding: A high like count does not provide useful information about a post's share count.

12. Likes vs Comments

Likes and comments also form a broadly unstructured scatter. Observations occur throughout the range without a discernible diagonal relationship.

The Pearson correlation is only 0.01, indicating a negligible linear relationship. Highly liked posts are therefore not necessarily highly commented posts.

Finding: Likes and comments represent distinct engagement behaviours in the observed data.

13. Engagement Correlation Matrix

The engagement correlation matrix quantitatively confirms the scatter-plot observations. Likes and shares have a correlation of approximately -0.00, likes and comments 0.01, and shares and comments 0.02.

All three relationships are effectively zero. This indicates that the engagement measures do not behave as interchangeable measurements of one underlying linear engagement pattern.

A post identified as successful using likes alone cannot automatically be assumed to be successful in shares or comments.

Finding: Likes, shares, and comments should be analysed as separate engagement dimensions rather than treated as equivalent measures.

14. Top 10 Users by Post Volume

The most active user contributed 22 posts, followed by users contributing 19 and 18 posts. The top group is relatively tightly clustered.

Across 12,000 posts and 1,500 users, the average is approximately 8 posts per user. Even the most active user contributes less than 0.2% of the dataset, indicating that no individual account dominates the overall post volume.

Finding: Posting activity is distributed across the user population rather than being concentrated in a small number of prolific accounts.

15. Follower Count vs Likes

The scatter plot shows a dense cloud across the follower-count range, with no visible upward relationship between audience size and likes.

The corresponding Pearson correlation is only 0.01. Accounts with larger follower counts therefore do not consistently receive more likes per post in this dataset.

Finding: Follower count is not a useful linear predictor of post-level likes within the observed data.

16. Posts by User Language

The language distribution is diverse. Chinese (zh) has the largest representation, followed by Japanese (ja) and Hindi (hi), with English (en) also forming a substantial group. The remaining languages occupy smaller but meaningful portions of the dataset.

No single language represents an overwhelming majority. The dataset therefore contains a materially multilingual user population.

The variable represents the language recorded for contributing users; it should not be interpreted as automatically detected language of each individual post.

Finding: The dataset has substantial multilingual representation, with particularly strong representation from East Asian and South Asian language groups.

17. Posts by User Location

The location distribution spans major cities across North America, Europe, Asia, the Middle East, South America, and Australia. Los Angeles, Munich, Shanghai, and Barcelona are among the more represented locations shown.

The counts remain relatively close to one another, so no single location dominates the displayed geographic distribution.

Finding: The contributing user base is geographically diverse rather than concentrated in a single region.

18. Distribution of Post Text Length

Most available post text is concentrated between approximately 100 and 150 characters, with the highest density around the 110–130 character range.

The distribution is relatively compact in its central region, with a small number of very short observations and a limited upper tail. These extreme values represent a small part of the dataset and should not be allowed to define the overall pattern.

Finding: Most posts have medium-length text, with relatively few observations at the extreme ends of the observed length range.

19. Text Length vs Likes

Likes are distributed across almost the entire 0–5,000 range for most text lengths. There is no visible pattern indicating that longer posts consistently receive more or fewer likes.

The correlation between text length and likes is approximately -0.00, indicating effectively no linear association.

Finding: Post length provides essentially no linear explanation for likes in this dataset.

20. Overall Correlation Matrix

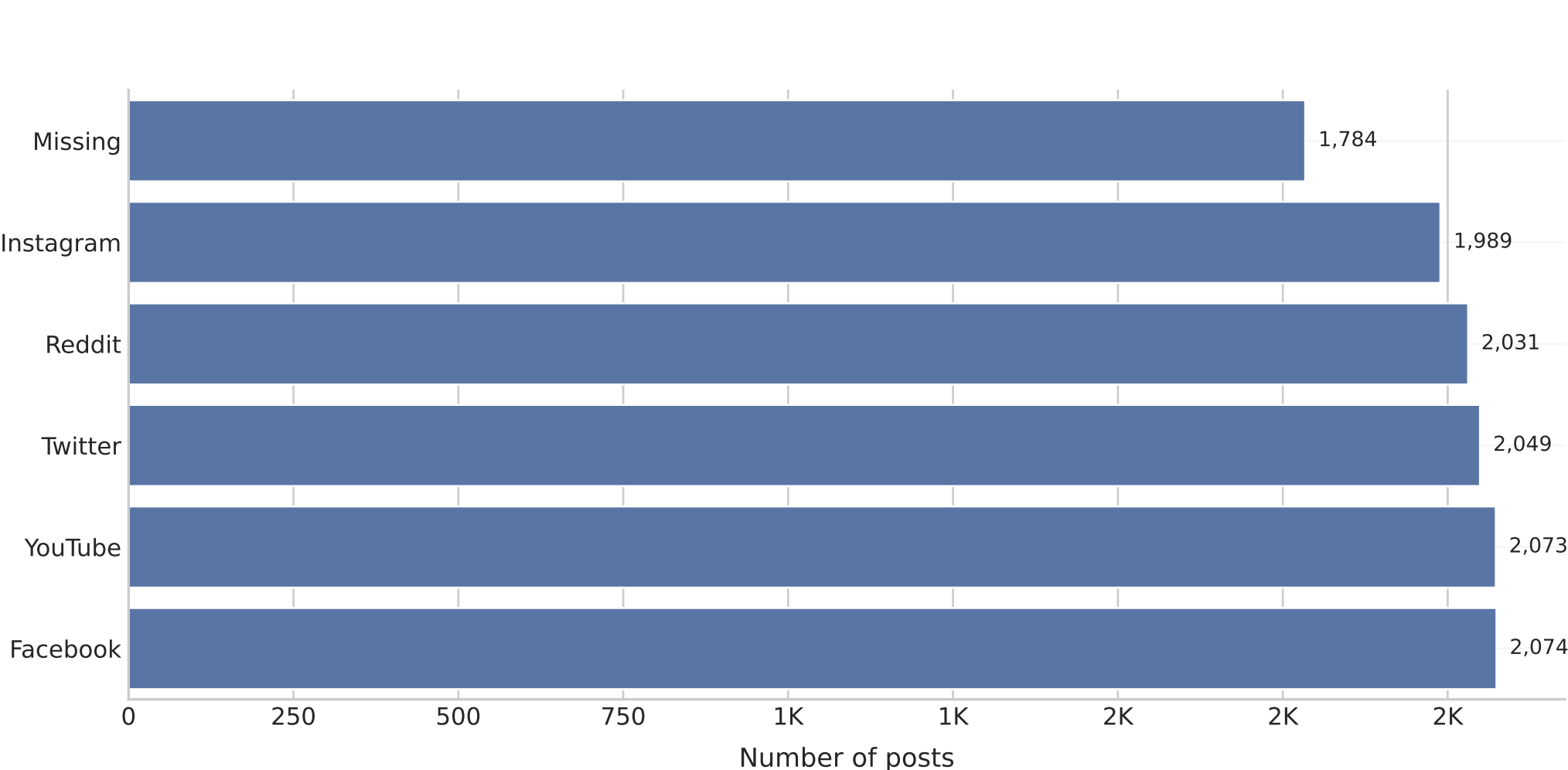
The full correlation matrix shows that the examined numeric variables are almost completely uncorrelated. The largest non-diagonal magnitude is approximately 0.03, between shares and text length.

Likes have correlations of -0.00 with shares, 0.01 with comments, 0.01 with follower count, and -0.00 with text length. The remaining pairwise relationships are similarly close to zero.

Finding: None of the examined engagement, audience-size, or text-length variables provides a strong linear explanation for another variable.

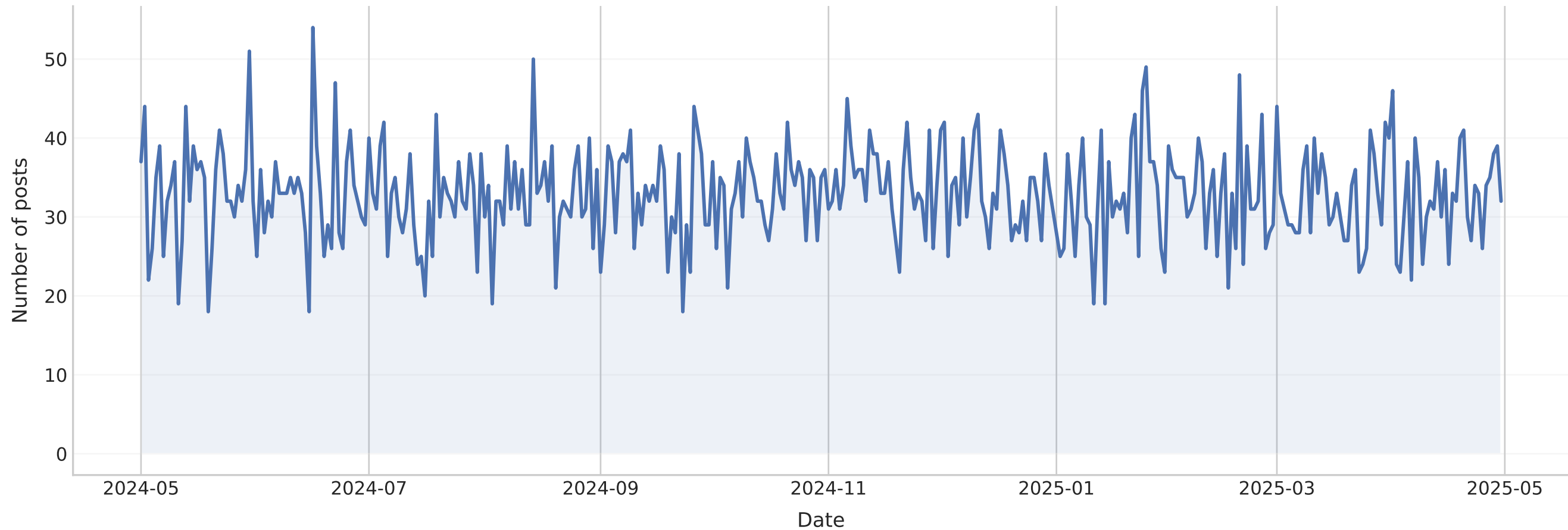
Post Volume by Platform

Distribution of cleaned posts across available platforms.



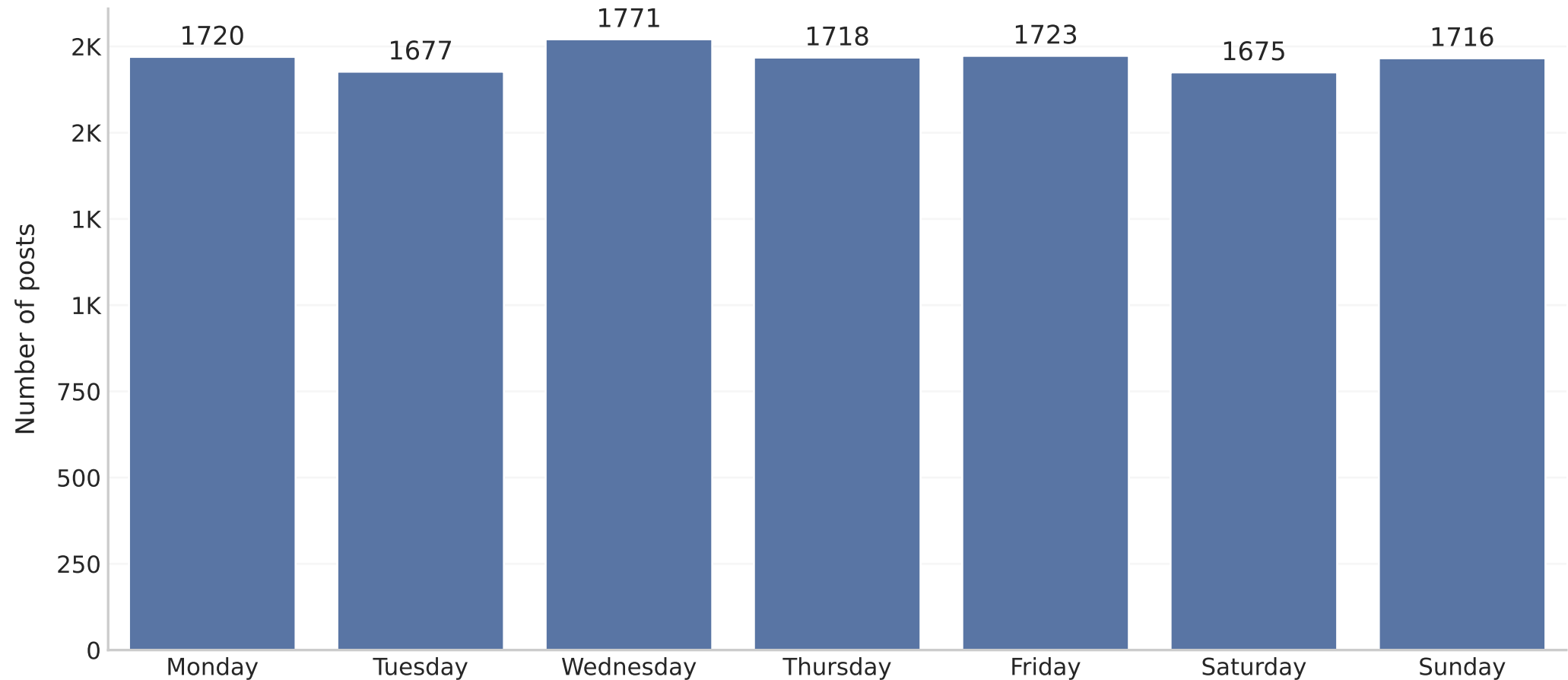
Daily Posting Activity

Daily post volume across the observation period.



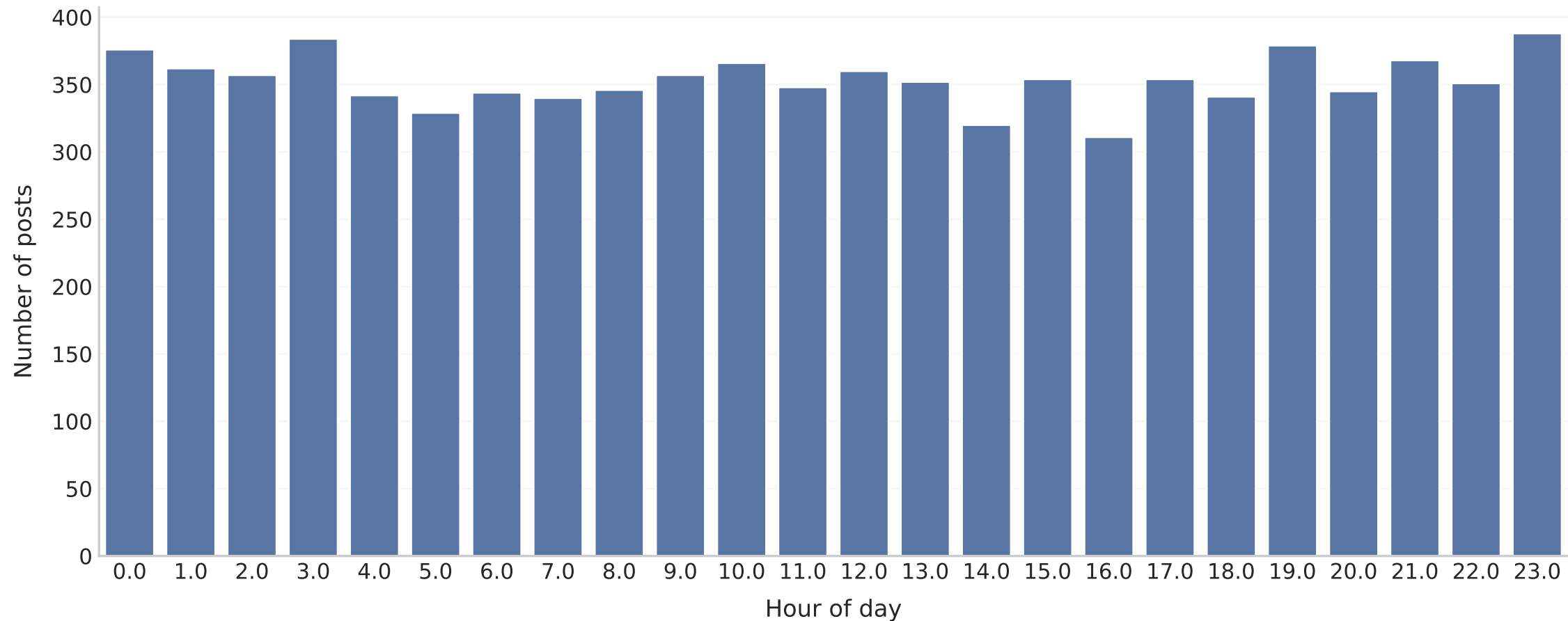
Posting Activity by Day of Week

Comparison of total post volume across weekdays.



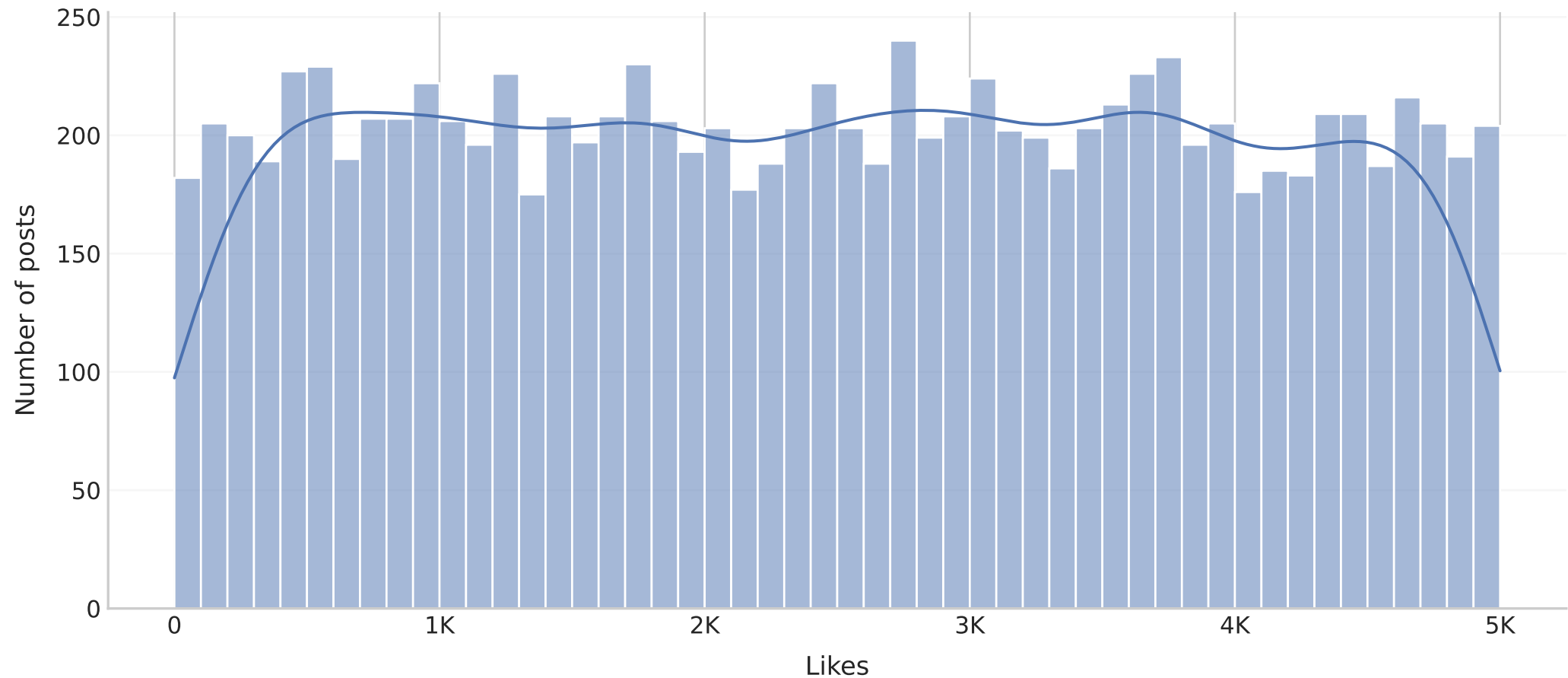
Posting Activity by Hour of Day

Only posts with a known posting time are included.



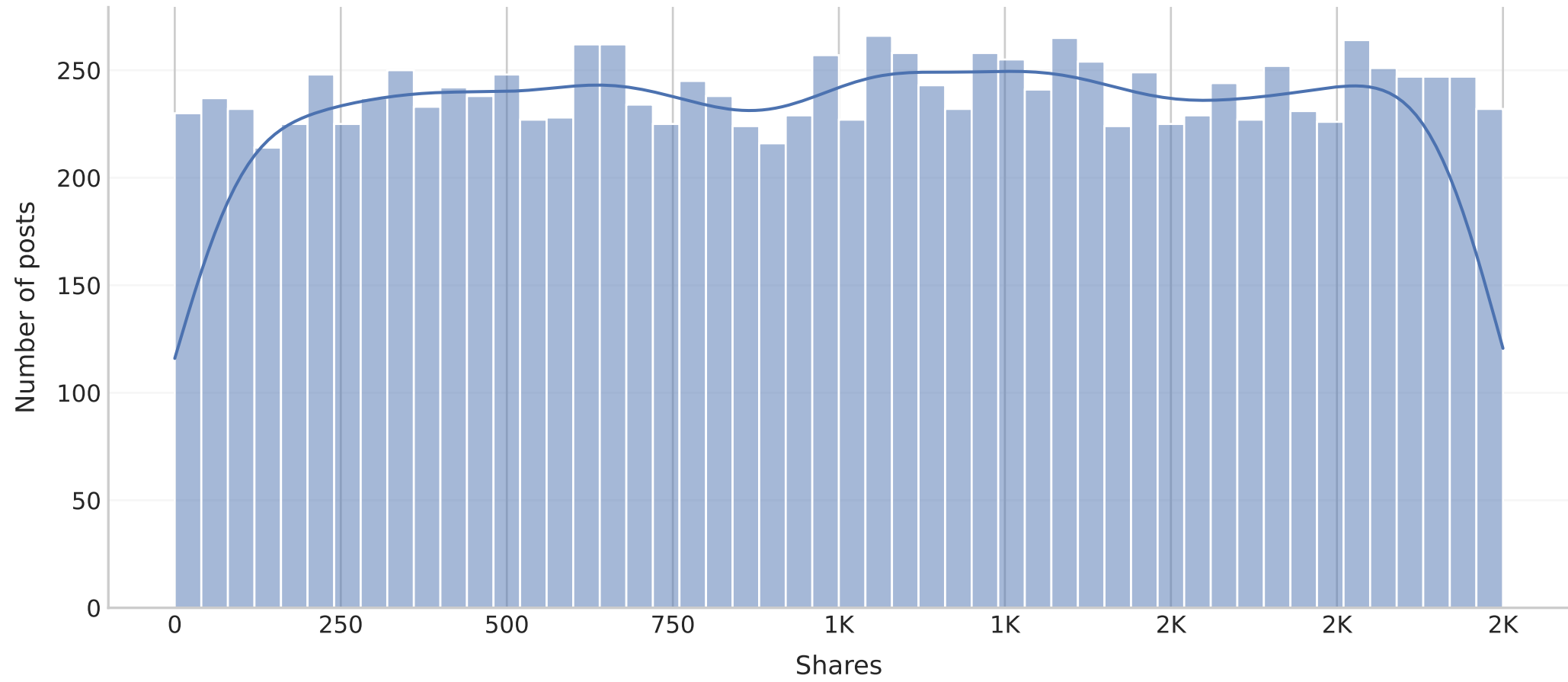
Distribution of Likes

Frequency distribution of post-level likes.



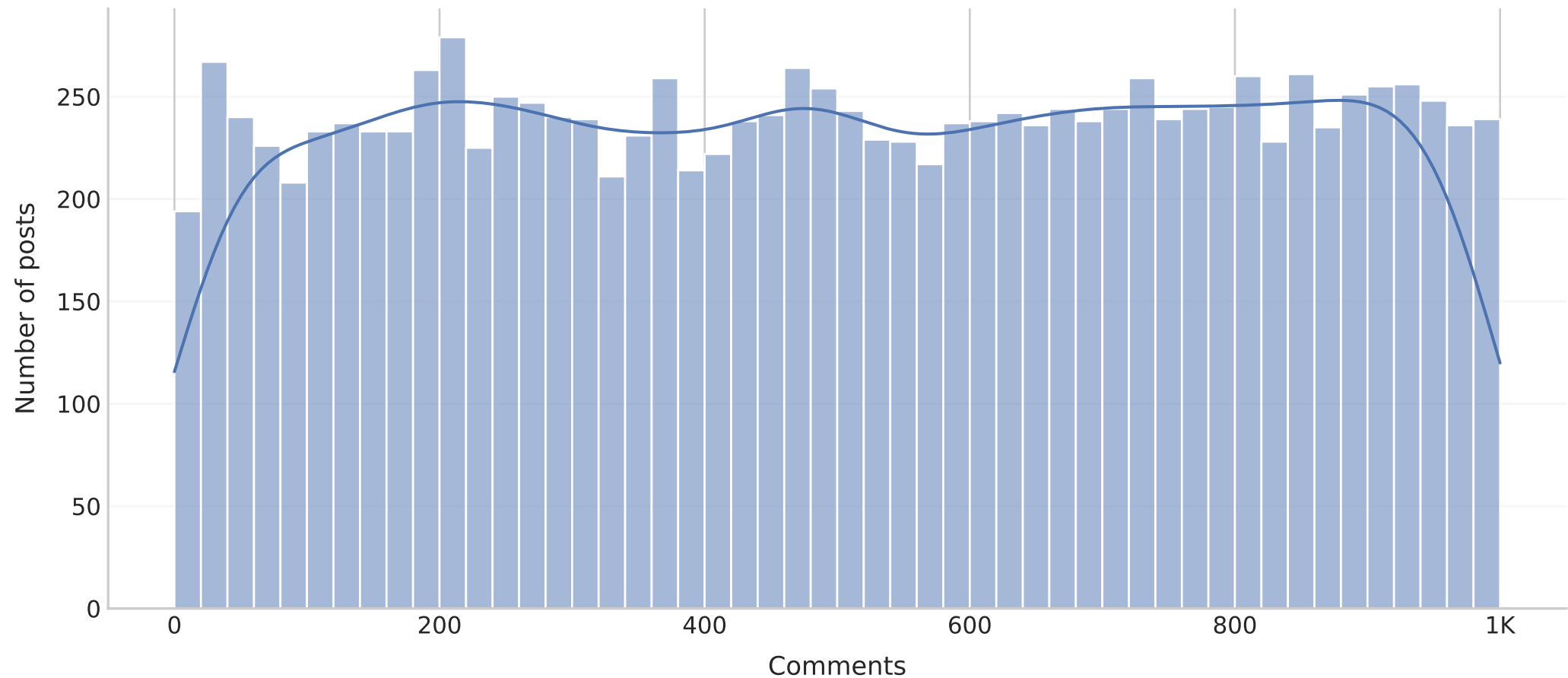
Distribution of Shares

Frequency distribution of post-level shares.



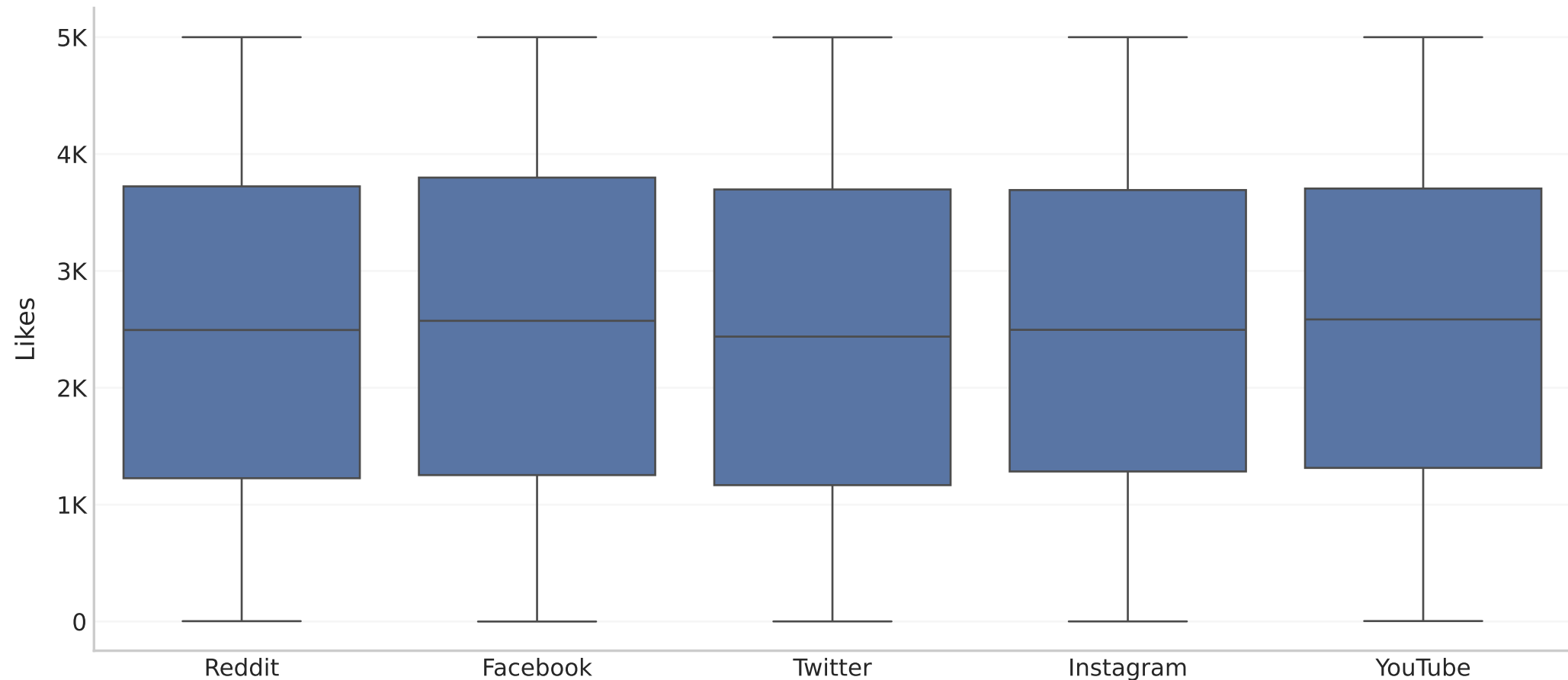
Distribution of Comments

Frequency distribution of post-level comments.



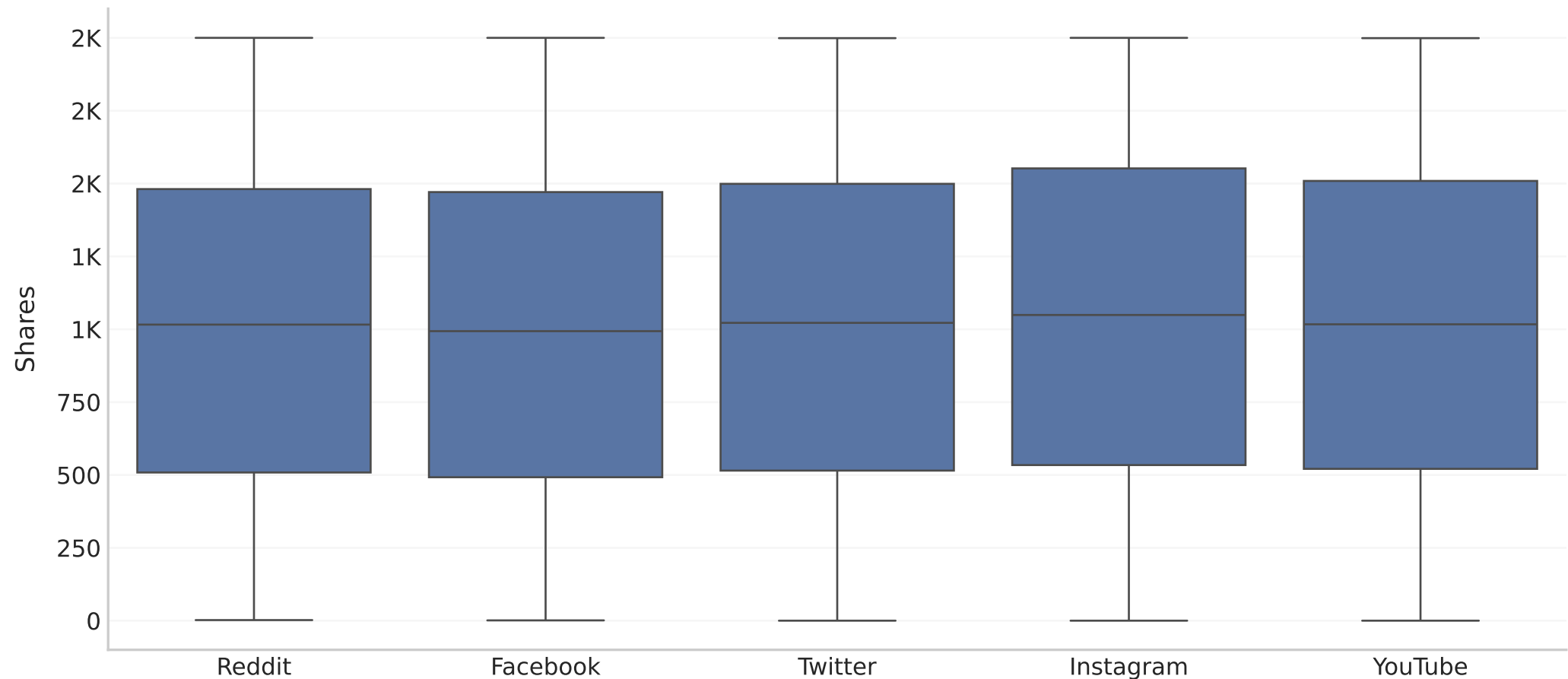
Likes by Platform

Box plots show the central tendency, spread, and extreme observations.



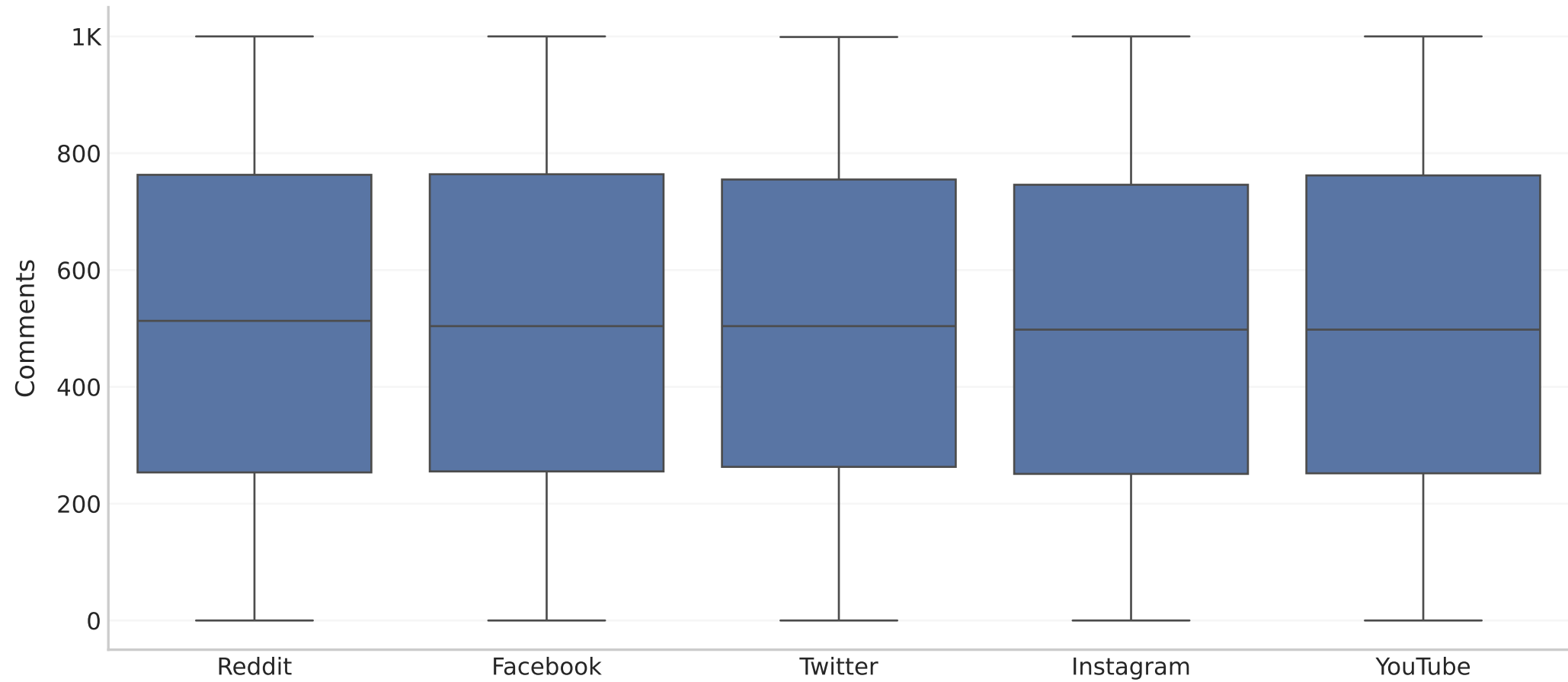
Shares by Platform

Box plots show the central tendency, spread, and extreme observations.



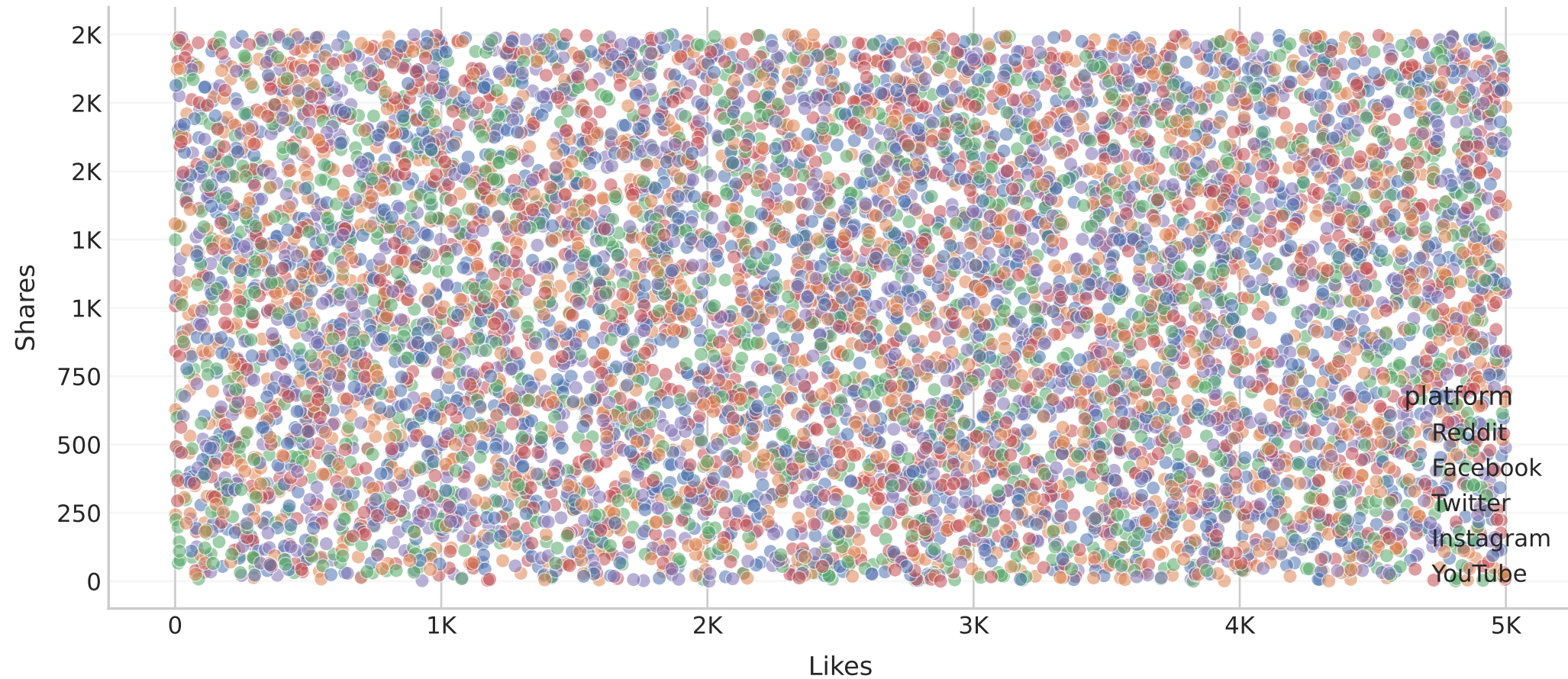
Comments by Platform

Box plots show the central tendency, spread, and extreme observations.



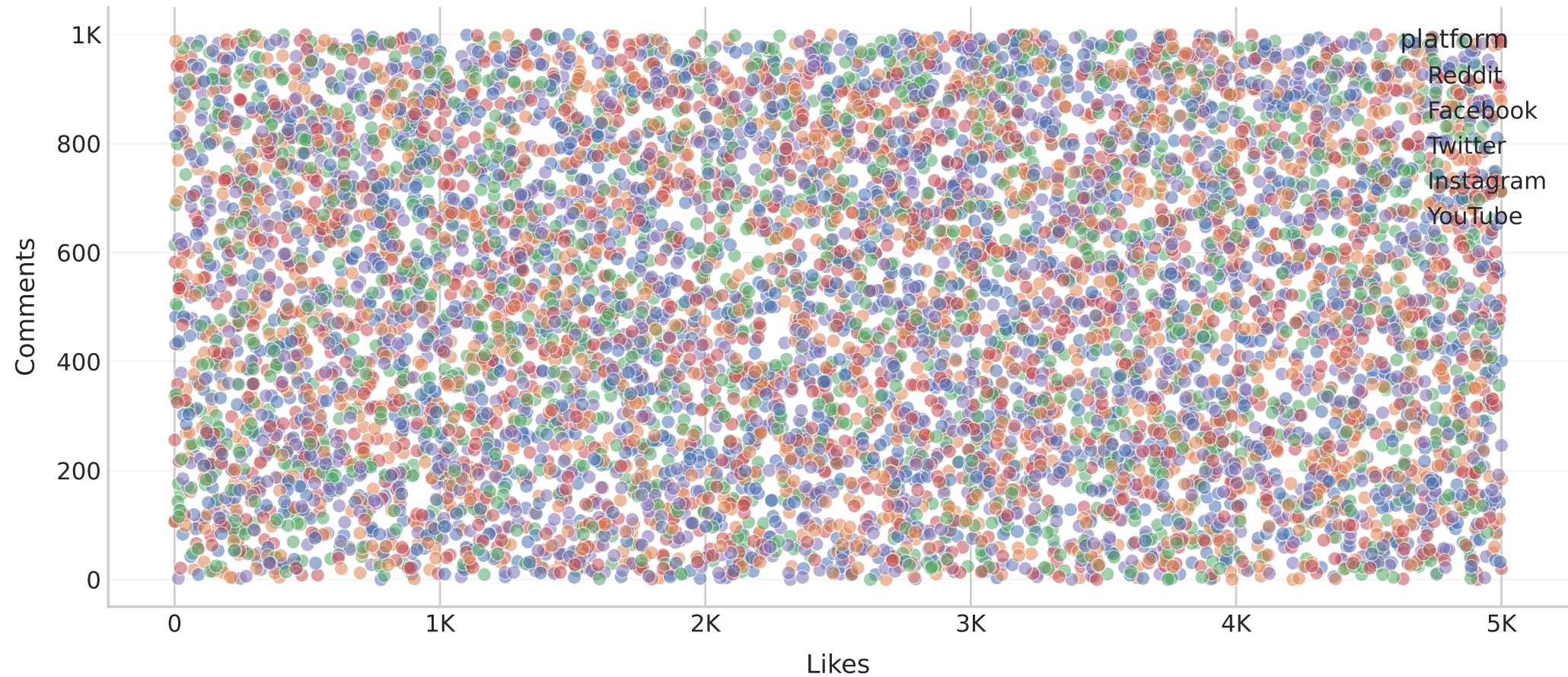
Likes vs Shares

Each point represents a post; platform is shown through grouping.



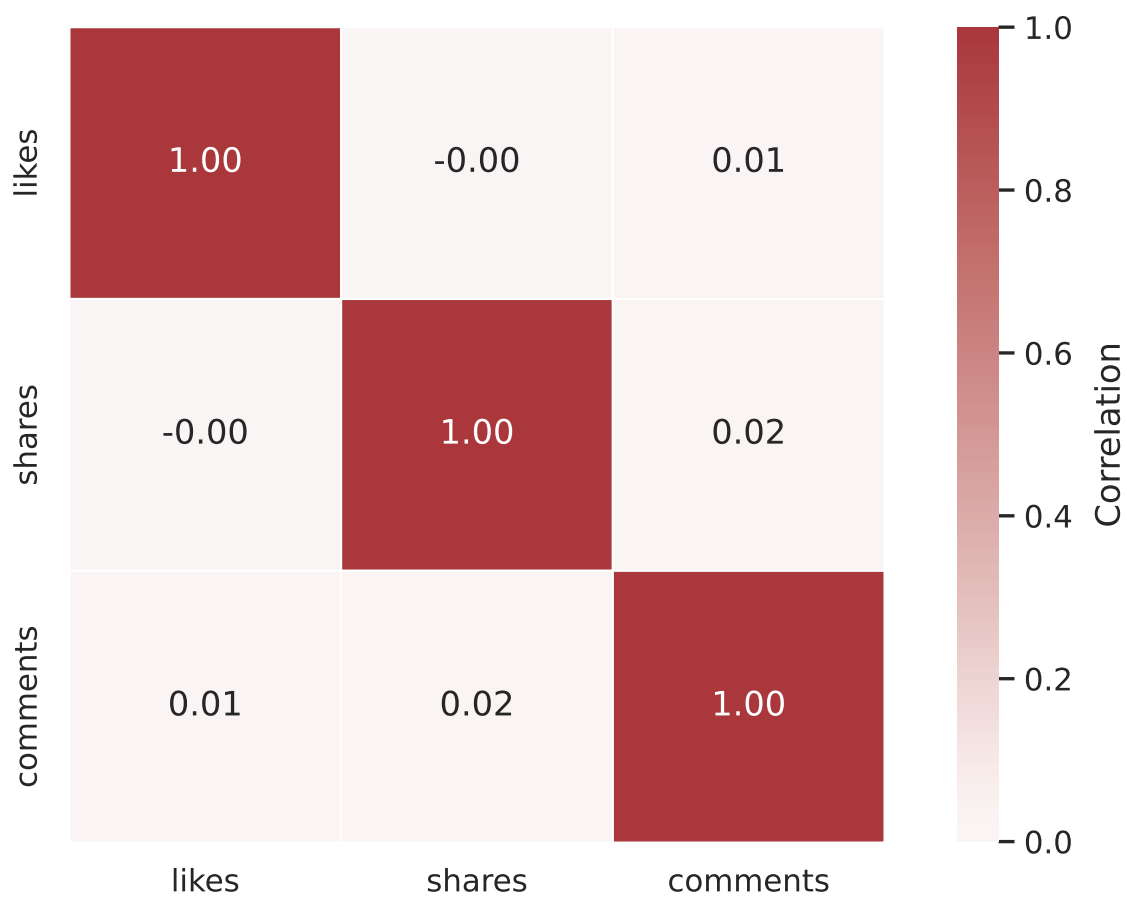
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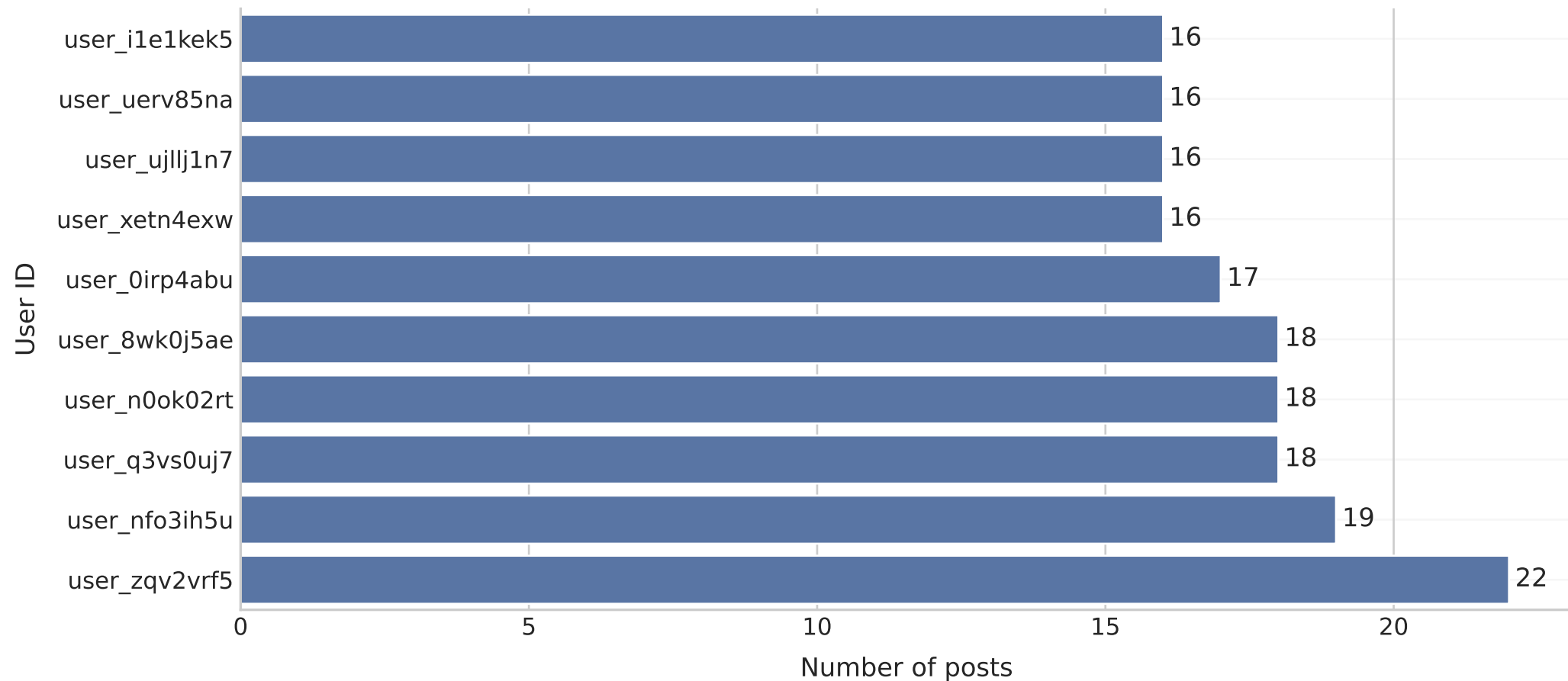
Engagement Correlation Matrix

Pearson correlation coefficients between engagement measures.



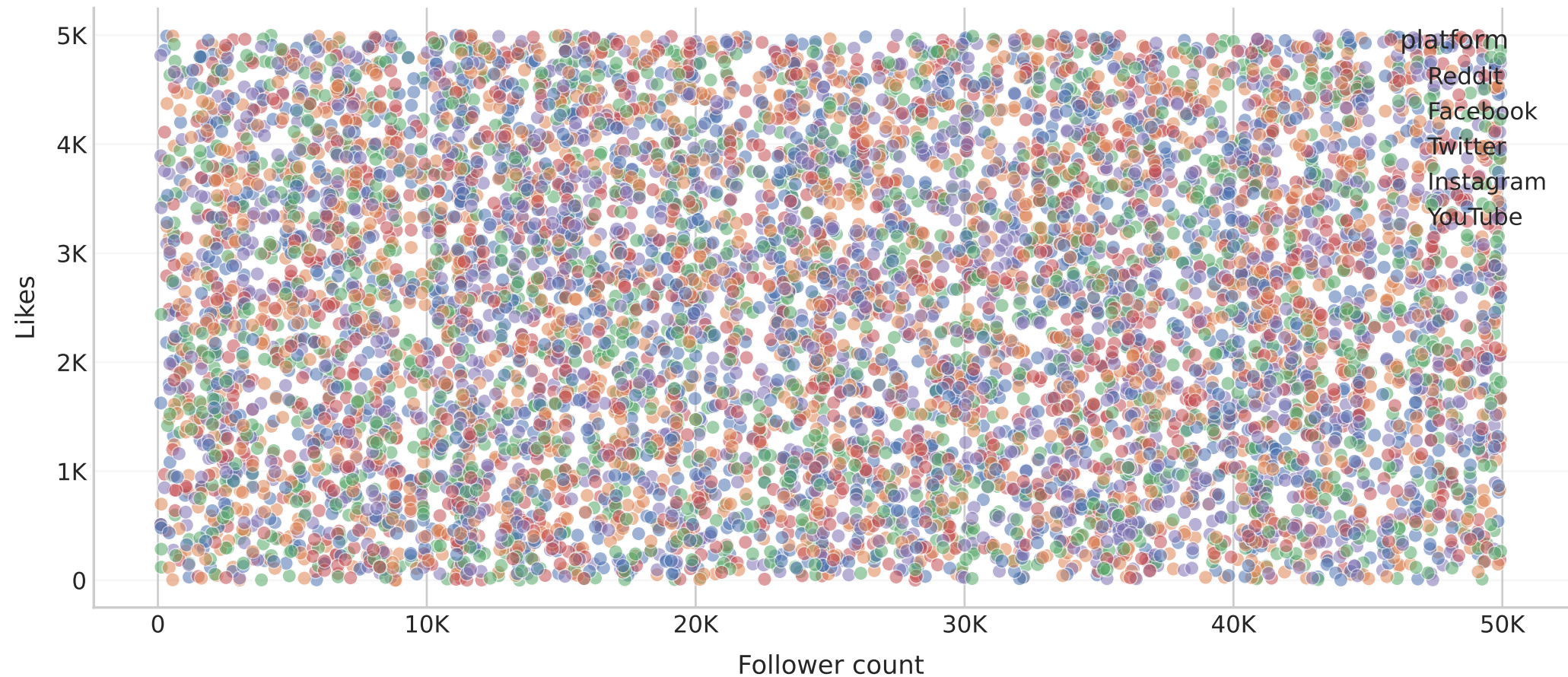
Top 10 Users by Post Volume

Users ranked by number of posts in the cleaned dataset.



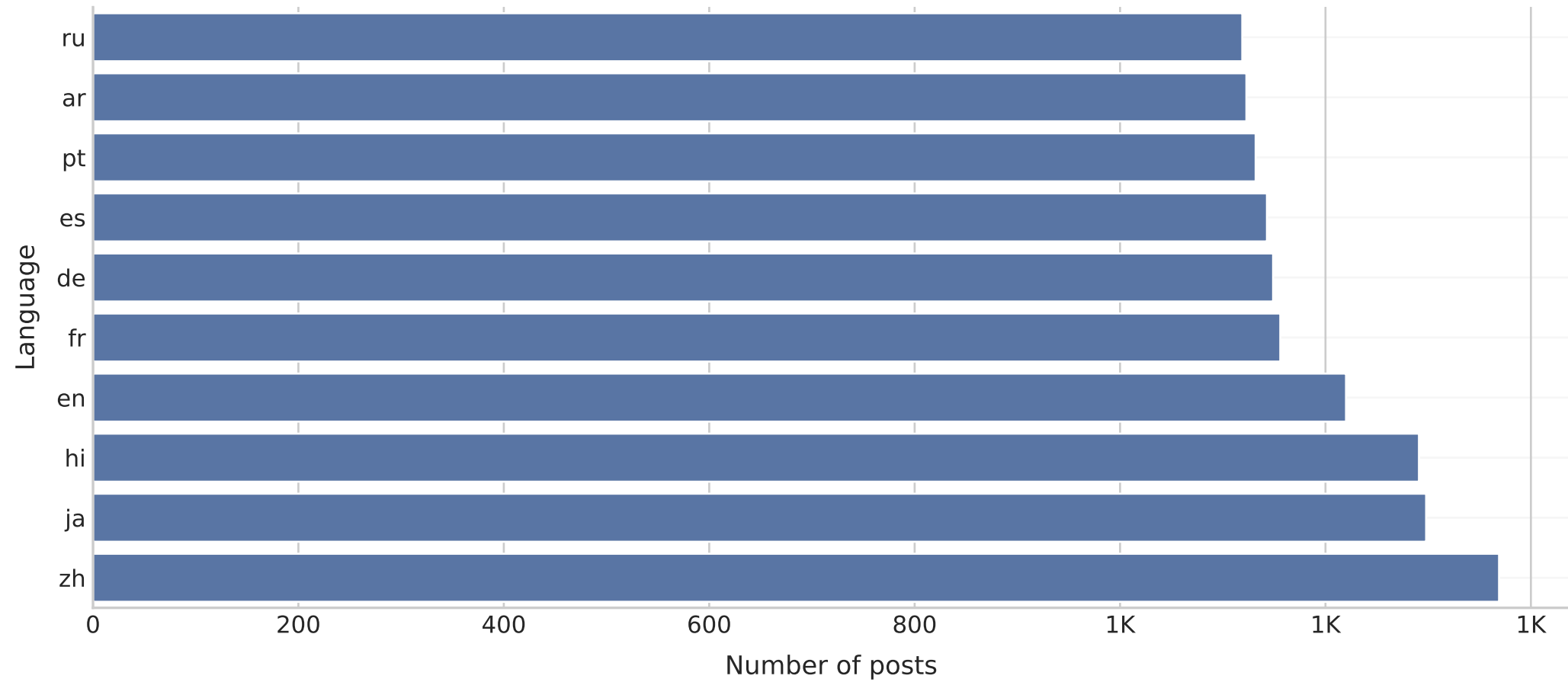
Followers Count vs Likes

Relationship between user audience size and post-level likes.



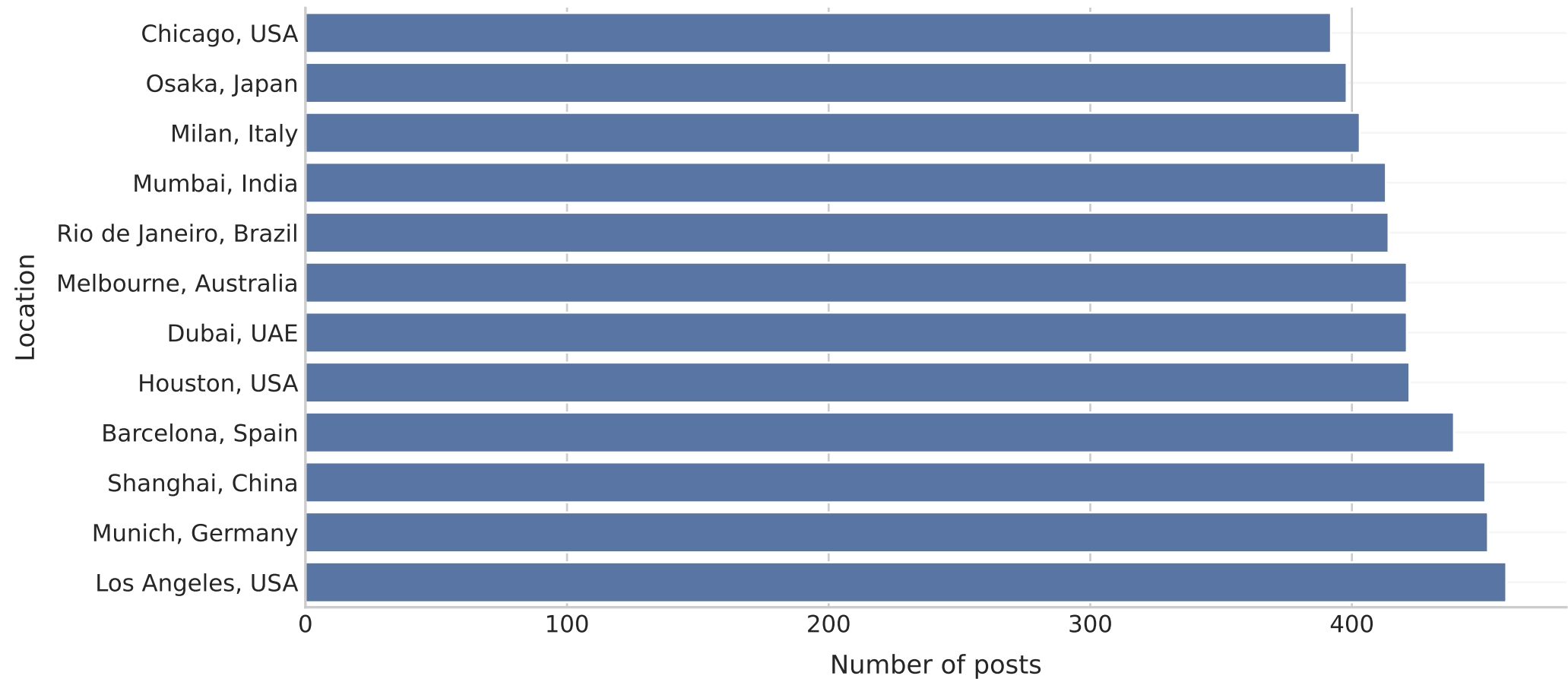
Posts by User Language

Top languages represented by users who contributed posts.



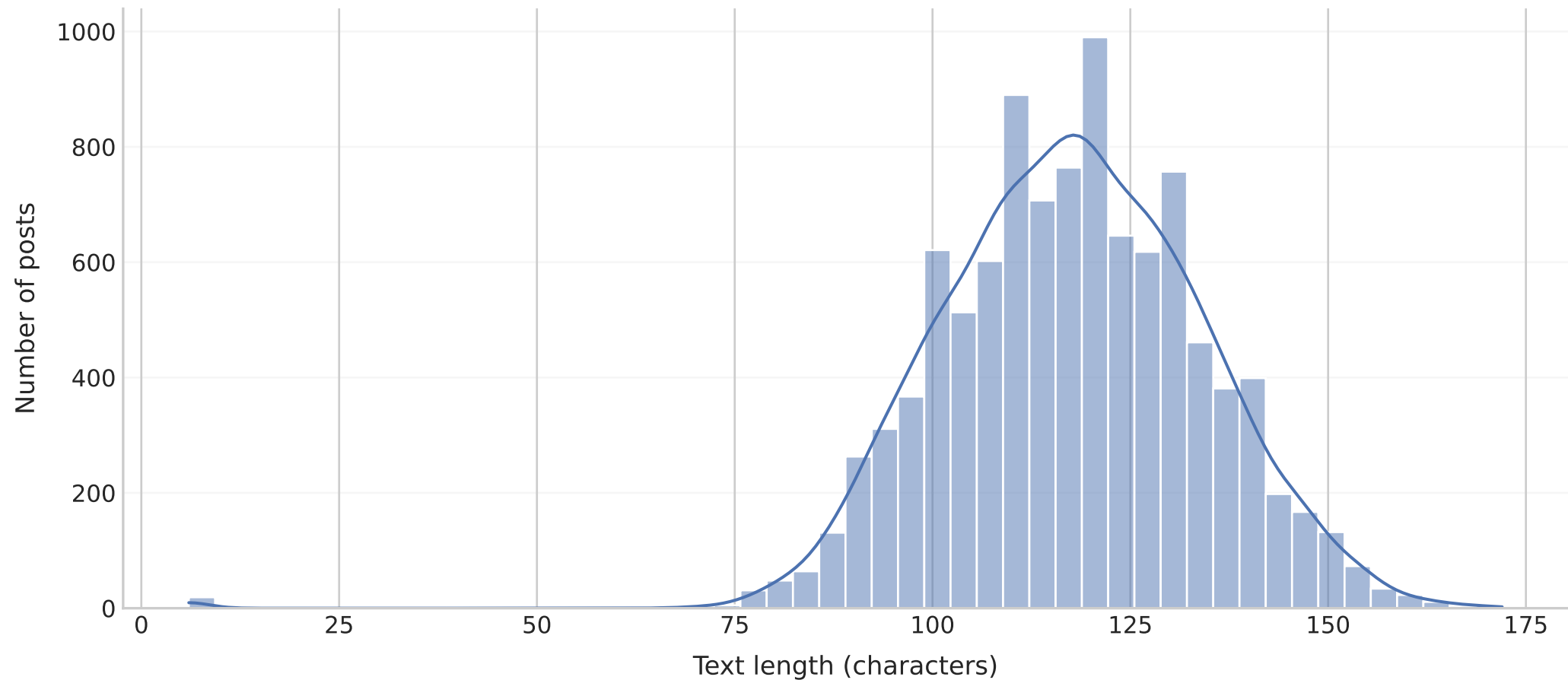
Posts by User Location

Top locations represented by users who contributed posts.



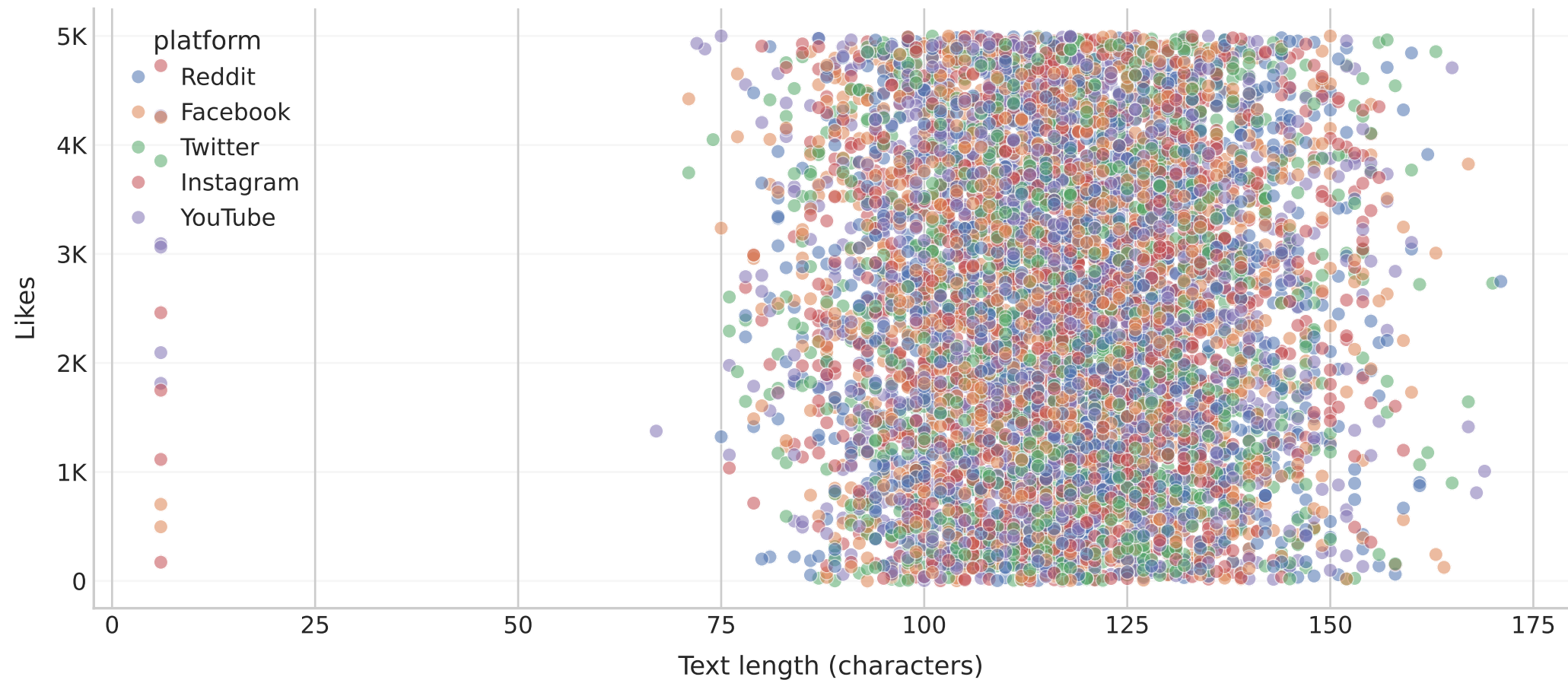
Distribution of Post Text Length

Character count after text cleaning; missing text remains missing.



Text Length vs Likes

Relationship between post length and likes.



Correlation Matrix of Numeric Variables

Pearson correlations between engagement, audience size, and text length.

